FDI Shocks and GDP Growth

Evidence from Global and Developing-Country Panels

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Introduction

Foreign direct investment (FDI) has been central to geopolitical and economic debate for much of the last century. Whether it be the United States encouraging private investment into Western Europe during the Cold War or China investing in Eastern Europe and Africa through its Belt and Road Initiative, FDI projects have played a role in shaping alliances and the economic order. Yet, while the geopolitical benefits to world powers are undeniable, the question remains: to what extent does FDI actually impact the recipient countries? This project seeks to answer that question by comparing FDI (as a percentage of GDP) with GDP per capita. Given that more investment should generally correlate with higher production, as well as FDI's general popularity, we hypothesize that FDI should positively correlate with GDP per capita.

Description of Econometric Models

We ran a number of regressions in which we varied different factors including the control variables, our definition of a treated country, and the model type. Our study of the relationship in question began with the simple bivariate linear regression below.

Data Summary

The primary data set for this paper comes from the World Bank Open Data set, where we found FDI (as a percentage of GDP), GDP per capita, and 13 other control variables for each country between 1960 and 2023. We also used the 2026 human development index values (HDI) from the World Population Review to create a list of "developing" countries. The table below features more information on the composition of our data set.

Table 1: Variable Definitions and Data Sources

Variable Name	Definition	Source
Total natural resources rents	As a % of GDP	World Bank Open Data
Control of Corruption: Estimate	Units in standard normal distribution	World Bank Open Data
Government Effectiveness: Estimate	Units in standard normal distribution	World Bank Open Data
Government Expense	Expenditures as a % of GDP	
Political Stability and Absence of Violence/Terrorism	An estimate of political stability and absence of violence/terrorism.	World Bank Open Data
Gini Coefficient	A measure of economic inequality. 0 = perfect inequality, 1 = perfect equality	World Bank Open Data
Tax Revenue	Tax revenue as a percentage of GDP	World Bank Open Data
Expense	Government expenditures as a percentage of GDP	World Bank Open Data
Rule of Law: Estimate	Aggregate and individual governance indicators for six dimensions of governance: (0-100)	World Bank Open Data
Regulatory Quality: Estimate	Estimated regulatory quality by country: (0-100)	World Bank Open Data
Literacy Rate	Percentage of the population ages 15 and above that is literate	World Bank Open Data
Human Capital Index Plus (HCI+)	(scale 0–325)	World Bank Open Data
Energy Use	Kg of oil equivalent per capita	World Bank Open Data
Trade	Trade as a percentage of country's GDP	World Bank Open Data
GDP Per Capita	GDP per capita in 2015 USD	World Bank Open Data
Foreign Direct Investment	Net inflow of FDI as a percentage of GDP	World Bank Open Data
Human Development Index (HDI)	Score from 0-1 based on UN metrics	World Population Review

The table below features the summary statistics for each variable.

Table 2: Summary Statistics for Key Variables

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
treatment_group2	2	0	0.20	0.40	0.00	0.00	1.00
GDP Per Capita (t+3)	4689	15	11965.54	15623.10	226.44	4598.39	81608.41
FDI (% of GDP)	4872	11	3.93	6.12	-82.89	2.57	103.34
Rule of Law	4517	16	-0.13	0.96	-2.59	-0.28	2.12
Energy Use (kg oil eq.)	3846	30	2308.97	2995.61	9.73	1165.61	21557.48
Trade (% of GDP)	4471	19	81.22	49.26	0.03	72.44	863.20
Natural Resources (% GDP)	4425	12	7.03	10.82	0.00	2.11	79.43
Tax Revenue (% GDP)	2923	47	16.95	17.75	0.00	15.80	638.66
Gov. Expense (% GDP)	2879	48	26.48	21.63	0.00	25.02	758.96
Corruption Control	4438	17	-0.12	0.95	-1.97	-0.33	2.46
Political Stability	4446	17	-0.12	0.99	-3.31	-0.04	1.96
Regulatory Quality	4443	18	-0.13	0.95	-2.55	-0.24	2.22

In the World Bank .csv file, each row measured one variable and one country pair over time. We used an R script to reconfigure that data set into traditional panel data, a much easier style to work with. We then created the graph below, which maps the percentage of missing values of each variable over time.

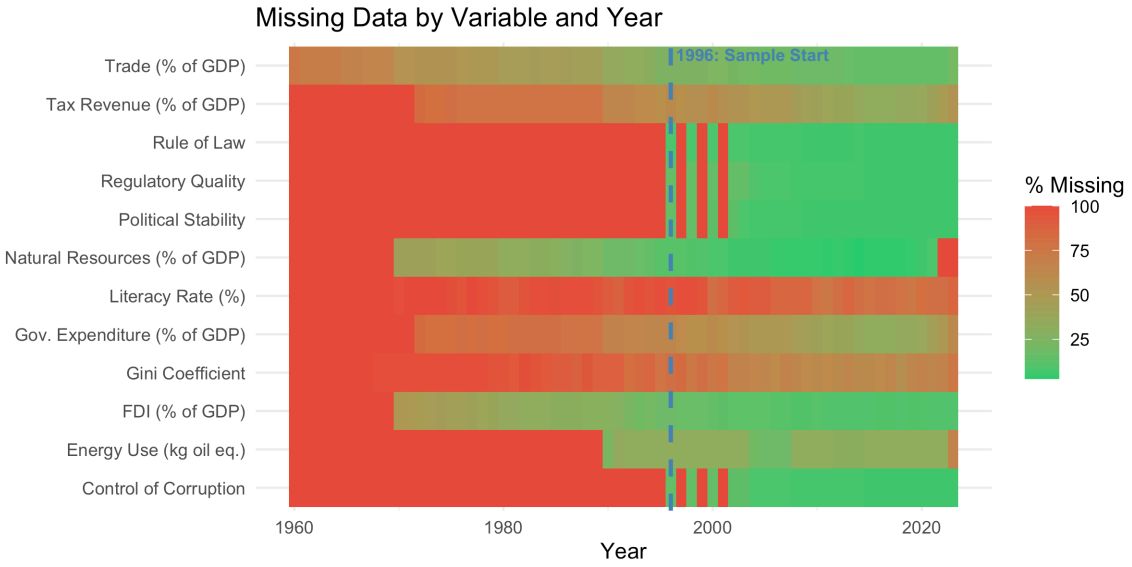


Figure 1: Missing Data Heatmap

Given that many of our variables had missing responses in the years prior to 1996, we chose not to consider the period between 1960 and 1996. While this meant we could not model some interesting geopolitical trends, such as the US FDI during the Cold War, the risk of false conclusions due to a lack of data was too high to keep those years in the dataset.

To determine which control variables were necessary for modeling FDI and GDP, we employed both statistical and logical processes. First, we manually ran a forward stepwise regression. To do so, we constructed a full linear regression model with each of the 13 controlling variables and FDI. We then removed the variable with the highest p-value, government effectiveness, and ran a new regression with the remaining variables. We repeated the process - removing the variable with the highest p-value and running the regression again - until each of the remaining variables was statistically significant at a basic level. The remaining variables were: natural resources rents, corruption index, Tax Revenue (% of GDP), Government expenditures (% of GDP), regulatory quality, energy use, trade, and FDI (% of GDP). This was a good starting point, but we did not want to solely rely on statistics to choose our control variables. A country's GDP is an incredibly complex function of countless different factors, so we wanted to examine it from a holistic and logical perspective. With the stepwise regression in mind, we chose predictor variables that we felt represented a variety of causes of economic growth. This list consisted of: `treat_post`, `energy_use`, `trade`, `nat_resources`, `tax_revenue`, `gov_expense`, `corruption`, `pol_stability`, and `reg_quality` to be our control variables.

To ensure our regressions hold economic value, we next considered our variables in light of the Gauss-Markov assumptions. Given that each regression is linear in parameters and we are sampling every country equally, assumptions one and two were trivially satisfied. For assumption three, we created the following graph depicting the collinearity between predictor variables.

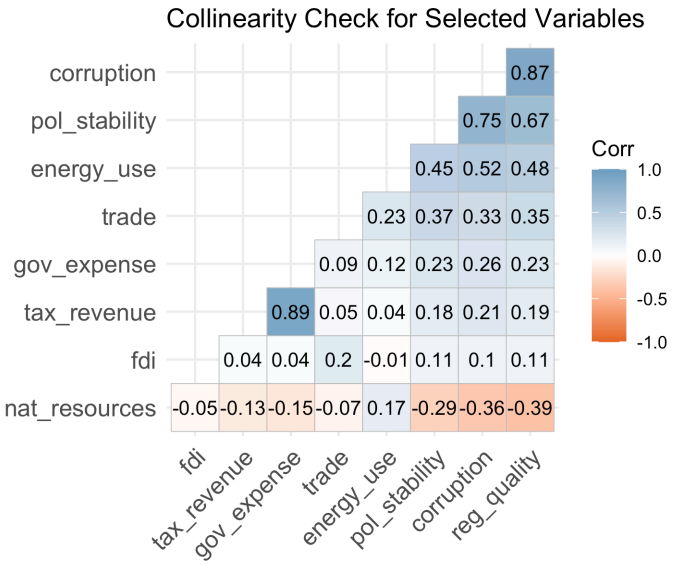


Figure 2: Collinearity Plot

By making a chart of each variable's correlation with each other variable, we can easily check the third Gauss-Markov assumption of no perfect multicollinearity. We find that we meet the criteria of no perfect multicollinearity, but there is some multicollinearity that we need to keep in mind going forward. To satisfy the 0 conditional mean assumption, we calculated the residuals from each model and then plotted them for different right side variables; in each case, we found that their mean was close to 0. We used clustered standard errors in each regression because there is a strong correlation in the error term between years for each country.

We reasoned that many of our variables would not impact an economy instantaneously but rather would take effect after a few years. To address this, we created 'GDP3', a statistic that measures the GDP per capita of a country three years in the future. After researching how long it took FDI to impact an economy, we decided three years is enough time for FDI's effect to manifest, but not be clouded over by other macroeconomic shifts. Additionally, we consulted some other professional economic papers on similar topics, and 3-5 years was standard practice.

Model Types

We used several types of models in our data analysis, and tested or considered even more that ultimately were not convincing enough to include. We began with some basic linear regressions before employing some more complicated techniques like TWFE and IPW DiD. The original basic linear models were primarily a tool to learn more about the data. We correctly predicted that the bivariate OLS would suffer from omitted variable bias while the full OLS model would likely overfit our data. As such, we were primarily interested in the model using only our selected variables. This would serve as our first glance test into whether high FDI levels could cause higher levels of GDP per capita. The regression table below features output from our first three linear regressions: the bivariate regression, the regression with all variables, and the regression with selected controls.

	Bivariate Model	Full Model	Specified Variables
(Intercept)	13015.281***	-301361.968***	11304.486***
fdi	79.330***		62.333**
rule_of_law		428.750	2162.884*
gov_effectiveness		540.548	

	Bivariate Model	Full Model	Specified Variables
corruption		-605.161	1845.634**
energy_use		1.750***	3.042***
reg_quality		-153.739	1663.329*
gov_expense		6.743	195.133***
literacy		-55.146+	
pol_stability		41.175	896.993*
nat_resources		66.158*	-139.622***
gini		52.266+	
tax_revenue		-7.806	-525.334***
trade		-0.755	-34.357***
year		152.955***	

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

As mentioned before, we ran the bivariate linear regression as an initial test to see if FDI would affect GDP3. While the independent variable was statistically significant, an R^2 value of .03 indicated that the simple model explained next to none of the variation in GDP per capita, as we expected. The full model had almost the complete opposite effect; it had many insignificant variables and quite a bit of overfitting. The third model only contained the control variables we listed earlier. This model had an R^2 of .8 and a significant, positive FDI coefficient. This was an interesting first finding, but it was far from enough to prove causality. Due to the complexity of everything that affects a country's GDP, we still assumed there was significant omitted variable bias in our model. As is, we have no control over shocks to the global economy, and our country's controls weren't nearly strong enough.

That assumption led us to the conclusion that a Two-Way Fixed Effects model could add Econometric validity to our model. The fixed effects we settled on were year and country. Including country as a fixed effect controlled time invariant differences between coun-

tries, such as geography and work culture. The time fixed effect would allow us to control for shocks that affected every country's GDP equally, such as COVID. The fixed effects combine to control many variables we couldn't possibly quantify or find adequate data on otherwise.

	(1)
treat_post	-845.165+
rule_of_law	895.674
energy_use	0.330
trade	12.599+
nat_resources	38.968*
tax_revenue	26.878
gov_expense	-21.075
corruption	-309.716
pol_stability	-255.395
reg_quality	1405.716*

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

After sorting the panel data by FDI net inflows, we found another problem: some countries had unreasonably high FDI to GDP ratios. We realized that our FDI data did not distinguish between actual investment in the local economy and accounting tricks employed by multi-national corporations to skirt tax obligations. We recognized some of the countries with the highest FDI, Luxembourg, for example were known tax havens. After cross-checking our data with a World Population Review report on tax havens, we decided to drop 25 countries from our data to ensure we were primarily looking at actual investment as opposed to accounting tricks.

While thinking about the application of our model we realized there may be some issues with how we are currently modeling FDI. FDI levels can fluctuate greatly year to year, meaning it is very easy for a country to receive a big FDI shock one year, exactly what we are interested in seeing the impact of, but then return to normal FDI levels in the following years. With the current model design, that one year of increased FDI would effectively get averaged out by all the other years. While normally a model's ability to moderate outliers is a good thing, in our case, it is those exact outliers we want to know the impact of. This led us to the solution that, rather than use FDI (% of GDP) as our variable of interest, we use a binary variable, treat_post. We defined treat_post to be 1 for a country at year t if at some point before or during year t the country had ever received an FDI that was between 1.5 and 2.5 standard deviations above the mean level. This means the country received an FDI inflow that was in the 93rd to 99th percentile. Countries without that property have entries of 0 for `treat_post`. This created a more effective variable for looking at the prolonged effects of FDI, as once the country receives the shock, we tell the model to continue treating it as a treated variable.

Even with the new treatment variable we still ran into two more problems with our TWFE model. Firstly, from a conceptual standpoint, we were trying to compare every country in the world to each other. This meant that if a developing country were to receive FDI, many of the countries we would be comparing it to were highly developed countries like the United States. This caused a problem as there will obviously be underlying differences between the two countries that our controls and fixed effects can't absorb.

The second issue was the prevalence of missing responses in our data set. There was a significant gap between the number of observations in our data set and the number of observations the TWFE was regressing over. We learned that R would only regress over a data point if it had a full set of observations, even if it was almost entirely complete. Since many of our data points were incomplete R was dropping them from its regression. After looking at the data, we found that some of our variables, government expense for example, were so sparse that the TWFE was only running on a small subset of the countries and years we intended. Even worse, we found that developing countries were significantly less likely to have full responses than developed ones. We reasoned that this would bias the TWFE into mostly considering the impact of FDI on already developed countries, which, while interesting, was not our research question.

Fortunately, both of these problems share a common solution, Inverse Propensity Weighting of the control units. By putting more weight on the countries that are most similar to the ones that received the FDI shock, we addressed the issue of comparing dissimilar countries across the model. The unexpected, but certainly appreciated, side effect was the fact that the IPW reduced the impact of our missing values. Since IPW relies on the averages of every variable to weigh the countries, within reason, missing values won't massively impact the IPW scores. Since we have considered the average values of all the variables for each specific country, we have effectively controlled for those variables in the regression even if we don't include them again in the model. This is why IPW models that include the control variables are considered doubly redundant. While this redundancy is normally

a very nice fail-safe to have, in our case, it is exactly what allows us to drop the control variables in our model that was forcing observations to be omitted. Our final model code for the global IPW TWFE model was:

$$\text{GDP}_{i,t+3} = \beta \cdot \text{treat_post}_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

The table below features the regression output for two IPW DiD regressions. In "Global" we regressed over all of the countries remaining in the data set after removing tax havens. In "Developing Countries" we only regressed over countries with a Human Development Index below 60 in 1990.

	Global IPW TWFE	Developing Countries IPW TWFE
treat_post	257.544	183.985
	(0.422)	(0.303)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Results

Our results are presented in rough order from most simple to the most complex, ending with IPW-weighed models. The bivariate model (FDI as the only predictor) shows that it is a statistically significant predictor of growth. Though with an R^2 of just .03, it explains very little of the variation on its own. The full model has a much higher R^2 of 0.8, but since it utilizes every predictor variable and has no fixed effects term, there is significant overfitting as well as omitted variable bias. The diagnostic plots for this model show some heteroskedasticity and right-skewed residuals, which motivated our use of clustered standard errors by country in subsequent fixed-effects models. The stepwise regressions show

similar issues, though the automated forward stepwise model has significantly less concerning diagnostics than the manual backward model.

The next evolution of our model involved the use of unweighted Two-Way Fixed-Effects. Our first TWFE model regresses `gdp3` on `treat_post` and the 10 selected control variables, with country and year fixed effects, and standard errors clustered at the country level. The global model uses 1,835 observations across 111 countries and 22 years. In the global model, `treat_post` has a coefficient of −845 with a p-value of 0.076, which is just barely statistically significant. This suggests that countries experiencing an FDI shock of between 1.5 and 2.5 standard deviations above the mean saw GDP per capita $845 lower than countries that did not, after controlling for observables as well as absorbing country and year fixed effects. The sign reversal from the bivariate model is noteworthy because the naïve positive association was primarily driven by country differences and OVB that are now accounted for. Only two explanatory variables, regulatory quality and natural resource rents, are statistically significant at the 5% level. This model has a within R^2 of 0.086, suggesting a lack of explanatory power. After applying inverse propensity weights, `treat_post` flips sign from negative to positive in the global and developing country models, but is still not statistically significant in any sample. The within R^2 for the global model drops to 0.001 (compared to its R^2 of 0.986, likely resulting from overfitting), meaning `treat_post` explains none of the within-country GDP variation once countries are properly weighted. This, coupled with the sign switching, suggests the estimated effect of FDI shocks on GDP is negligible.

The primary identifying assumption of the TWFE approach is that both treated and control countries would have followed parallel GDP trajectories in the absence of an FDI shock. The parallel trend plots compare average log GDP per capita over time for both groups. The slopes are extremely similar. The lack of a divergence after treatment takes place is consistent with the findings from our regressions indicating a lack of effectiveness of FDI shocks on GDP.

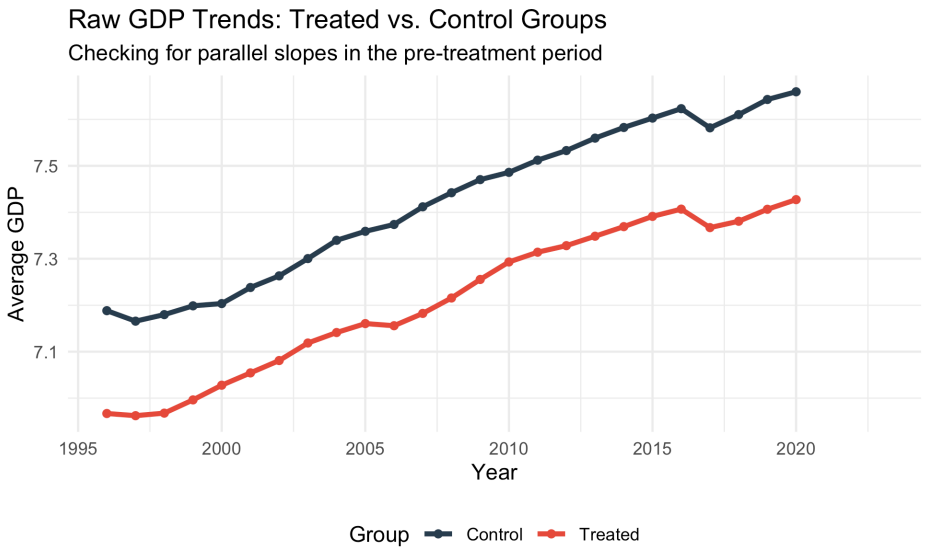


Figure 3: Parallel Trends: Global Treated vs. Control

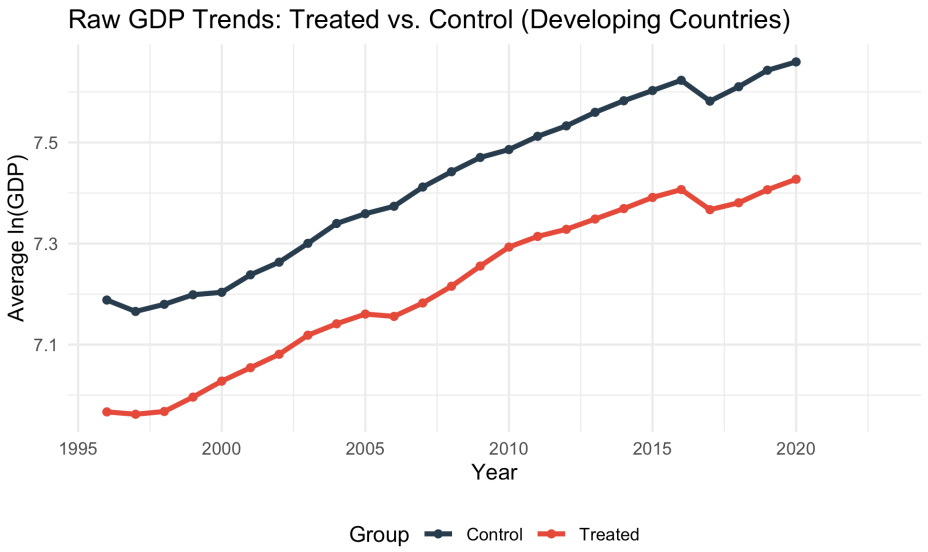


Figure 4: Parallel Trends: Developing Countries Treated vs. Control

Of the 170 countries in the global sample, only 40 experienced an FDI shock meeting our 1.5-2.5 standard deviation threshold, with just 13 of those countries in the developing country subsample. The estimated IPW coefficients of $256 and $169 are approximately 2 and 8 percent of their respective sample means for GDP per capita. Despite FDI's lack of statistical significance, the consistency of a null effect across every non-bivariate model and country group we studied suggests that this finding is genuinely a negligible effect as opposed to an inconclusive one or a lack of statistical power.

Discussion

We considered the causal impact of FDI on the GDP per capita of recipient countries. Our hypothesis, based on the assumption that a capital injection will increase output, was that an increase in FDI would cause GDP per capita to rise.

To test our hypothesis, we created a panel dataset primarily drawn from World Bank Open Data for the period between 1960 and 2023. After noticing that there was not enough

data prior to 1996, we elected to only model over the period 1996-2023. We also noticed that many known tax havens had unreasonably high FDI scores, so we dropped them from the data set to ensure we only measured genuine economic investment. We also transitioned from the continuous FDI (% of GDP) variable to the binary `treat_post` variable, defined as an FDI influx between 1.5 and 2.5 standard deviations above the historical mean, to ensure we could track FDI's effect over time.

We employed four models: bivariate OLS, full OLS, two-way fixed effects (TWFE), and Inverse Propensity Score Weighting TWFE. The bivariate OLS predicted a statistically significant positive relationship between FDI and GDP, but an R^2 value of just .03 indicated that the model had no explanatory value. The full OLS model had a much higher R^2 value and also predicted FDI to have a significant positive effect, but it suffered from significant overfitting. Realizing that an OLS was insufficient to understand the relationship, we next employed a TWFE model controlling for both time and country. This model could control for time-invariant qualities in each country as well as shocks to the global economy. Unlike in the OLS models, in the TWFE model, FDI had a negative coefficient and was not statistically significant. Unfortunately, we realized the TWFE model would only regress over a country-year pair if it had a full set of data entries. We employed an IPW DiD model as it limited the effect of missing data. This model only confirmed the TWFE's findings; however, FDI was still not statistically significant.

Our results were limited for a few reasons. Missing entries likely play the largest role in constraining our results. We could only regress over 1996-2023, given the lack of earlier

data. We also found that there was persistent missing data among developing nations, the exact nations we intended to study. While we were able to mitigate the impact of missing data through the IPW DiD, it still limited the precision of the propensity scores themselves. A final limitation is that FDI data has a lot of noise. Damgaard, Elkjaer, and Johannesen (2023) estimate that up to 40% of FDI ends up in corporate shells with no link to the real economy. Though we tried to control for this by dropping countries in which this is known to happen frequently, we could not account for all of it; our models, therefore, struggled to discern between real FDI and global accounting practices.

Today, leaders of developing countries pass policies to attract FDI. Our paper challenges whether doing so will have the benefits those leaders predict. We find that attracting an influx of FDI is unlikely to have any effect on a nation's economic trajectory. If anything, our findings support that the strong institutions have a larger positive impact on an economy than the amount of FDI it receives.

Appendix

	Global	Africa
treat_post	−845.17+ (472.50)	−101.42 (83.32)
Rule of Law	895.67 (586.66)	−126.48 (227.49)
Energy Use (kg oil eq.)	0.33 (0.26)	0.51 (0.31)
Trade (% of GDP)	12.60+ (6.45)	3.66 (2.87)
Natural Resources (% of GDP)	38.97* (17.14)	11.74** (4.01)
Tax Revenue (% of GDP)	26.88 (30.09)	−4.10 (5.17)
Gov. Expenditure (% of GDP)	−21.08 (27.88)	4.44 (4.71)
Control of Corruption	−309.72 (504.08)	130.24 (147.82)
Political Stability	−255.40 (242.32)	122.61 (88.02)
Regulatory Quality	1405.72* (579.07)	140.85 (205.10)
Num.Obs.	1835	363
R2	0.994	0.989
R2 Adj.	0.994	0.987
R2 Within	0.086	0.160
R2 Within Adj.	0.081	0.132
FE: country	X	X
FE: year	X	X

	Global (IPW)	Developing (IPW)	Africa (IPW)
treat_post	257.54 (319.76)	183.99 (176.80)	−59.00 (74.40)
Rule of Law			−152.23 (228.11)
Energy Use (kg oil eq.)			−0.24 (0.71)
Trade (% of GDP)			6.48** (1.83)
Natural Resources (% of GDP)			6.37 (4.61)
Tax Revenue (% of GDP)			1.47 (8.03)
Gov. Expenditure (% of GDP)			−1.87 (8.14)
Control of Corruption			243.84* (113.92)
Political Stability			103.81 (91.41)
Regulatory Quality			71.84 (130.94)
Num.Obs.	4656	1299	363
R2	0.982	0.974	0.986
R2 Adj.	0.981	0.972	0.983

	Global (IPW)	Developing (IPW)	Africa (IPW)
R2 Within	0.001	0.033	0.243
R2 Within Adj.	0.001	0.032	0.218
FE: country	X	X	X
FE: year	X	X	X

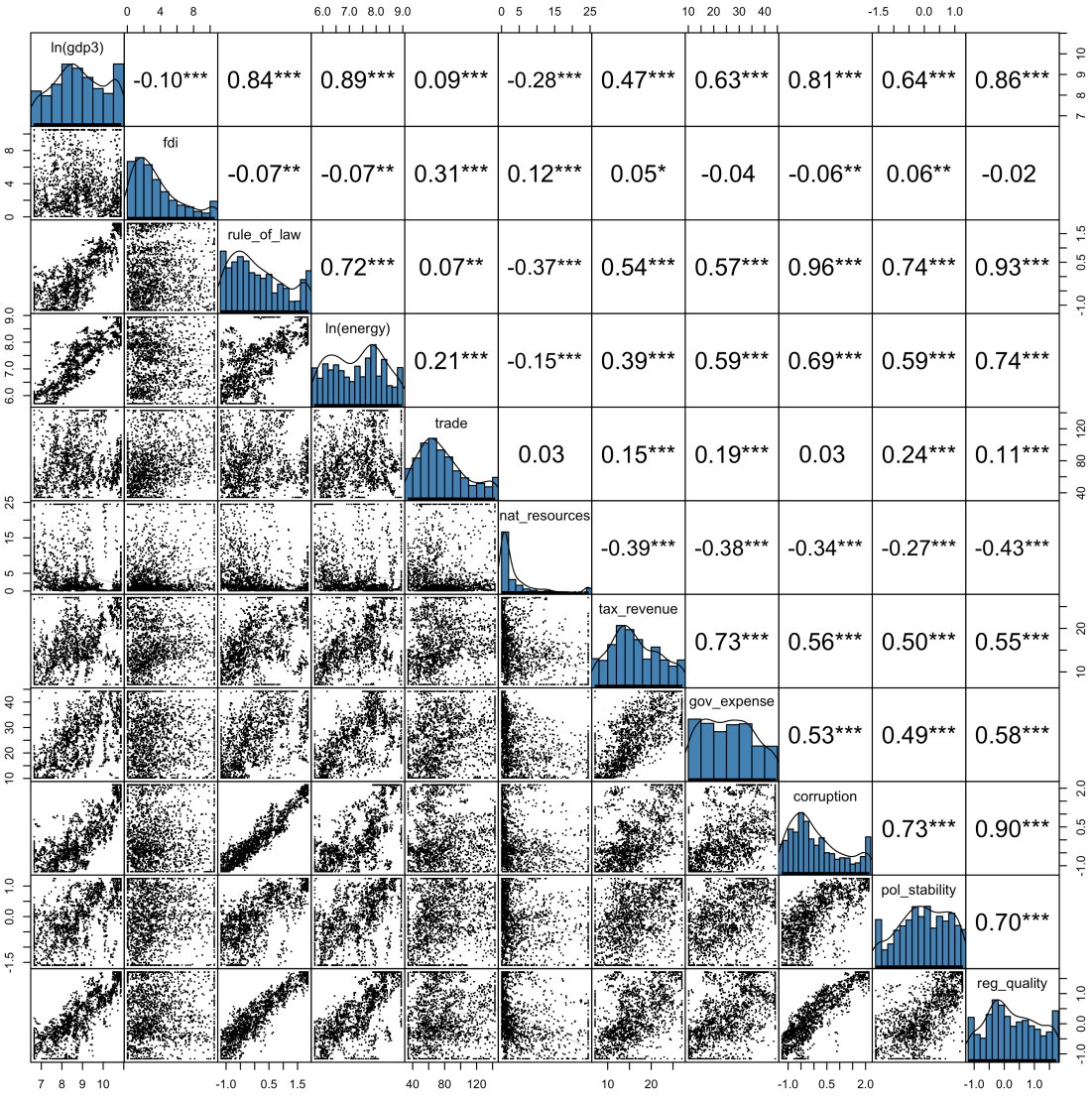


Figure 5: Pairwise Correlations and Distributions of Key Variables

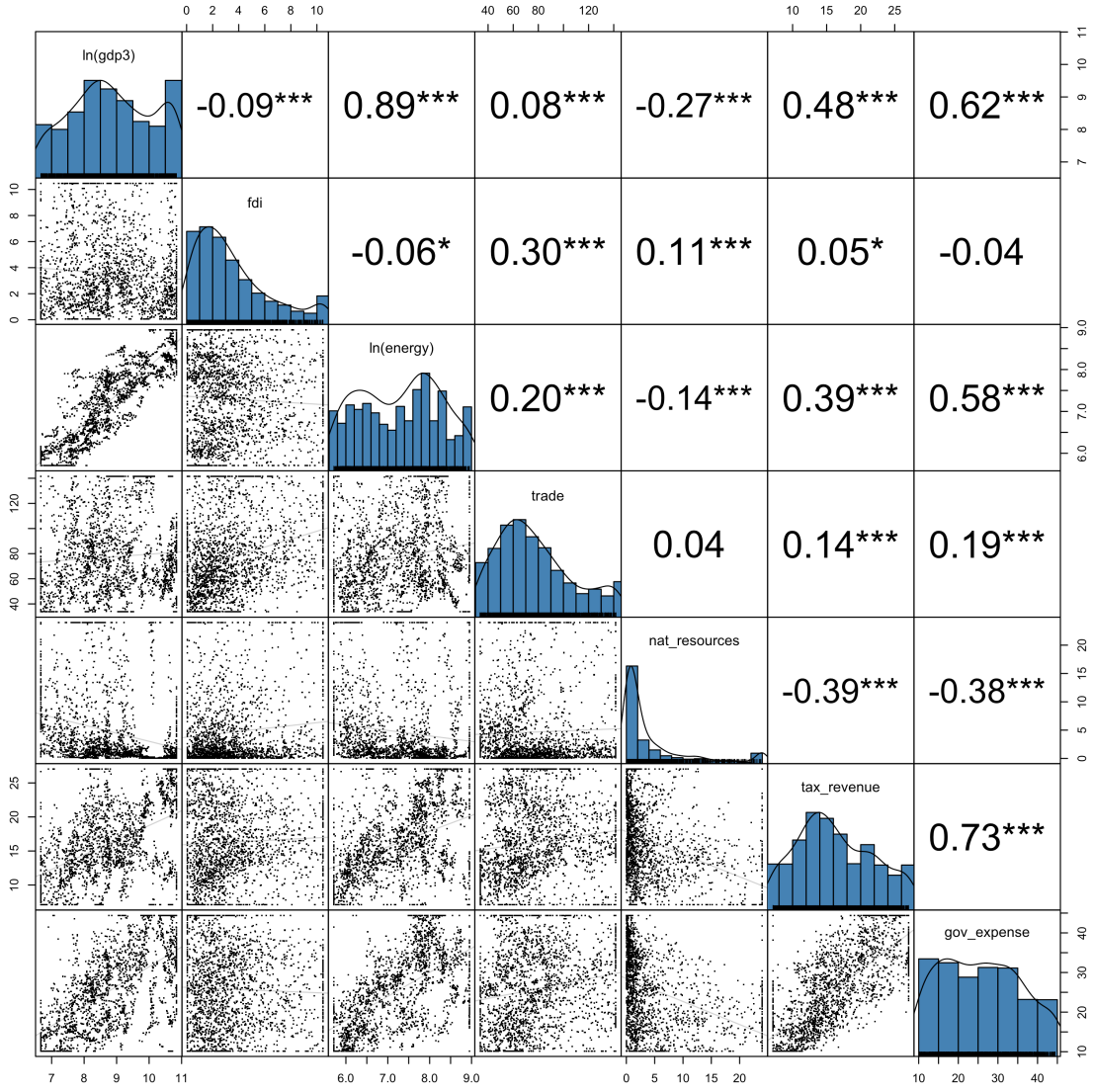


Figure 6: Pairwise Correlations of Economic Variables

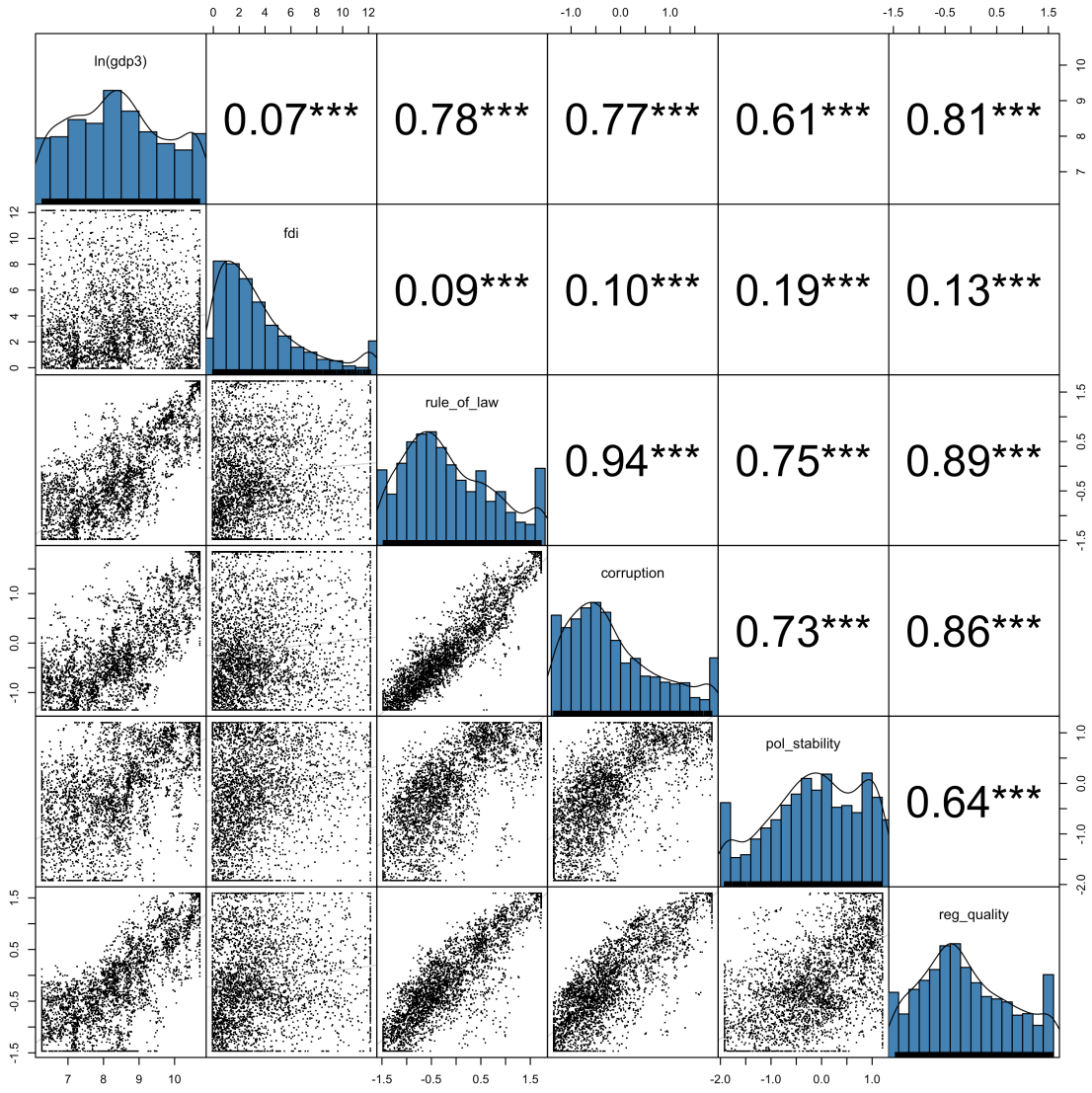


Figure 7: Pairwise Correlations of Institutional Variables

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